



MCA Semester - II (Year 2024_25)

UNIT : 1

Subject:- Full Stack Development

- 1 Describe the HTTP request and response cycle with an example.**
- 2 Differentiate between Static and Dynamic websites.
- 3 Explain the structure of an HTML document with an example.
- 4 Write an HTML code snippet to create a basic webpage with a form.**
- 5 Explain different types of CSS selectors with examples.
- 6 What is CSS Flexbox? Explain its properties with an example.**
- 7 Explain the event handling mechanism in JavaScript with an example.
- 8 Explain AJAX with GET and POST requests with an example.**
- 9 Explain different data types in JavaScript with examples.



MCA Semester - II (Year 2024_25) - Assignment

Subject Name: Full Stack Development (MC01094041)

Unit-2

- 1. Define Node Package Manager (NPM). Explain the process of installing, upgrading and uninstalling module using NPM.**
2. Write a Node JS code to create a server that will write a log of visited pages in a log.txt file.
- 3. How to create a simple server in Node.js that returns Hello World?**
4. What is Node.js? Explain the features of Node.js.
5. Explain how to serve files using the http module in Node.js with an example.
6. What is Express.js? Explain its features and advantages over the http module in Node.js.
7. What are templating engines? Explain with an example of using a templating engine like EJS, Pug, or Handlebars in Express.js.
- 8. Discuss the concept of static files in Express.js and explain how to serve static files in Express.js**
9. Explain async/await in JavaScript. How is it useful for handling asynchronous operations in Node.js?
- 10. Explain the working of the fetch() API in JavaScript and how it interacts with an Express.js backend.**



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UNIT : 3

- 1. How to create a simple server in Node.js that returns Hello World?**
- 2. Write code to perform Insert and find operations on Student database using Node.js and MongoDB.**
3. Explain usage of NODE_ENV?
4. Write code to perform update and delete operations on Employees database using Node.js and MongoDB.
- 5. Discuss NPM (Node Package Manager) in Node JS.**
6. What is MongoDB? Describe its architecture and core components.
7. What is request body parsing in Express? Explain with an example.
8. Compare NoSQL (MongoDB) and SQL databases with advantages and use cases.
- 9. What are cookies in Node.js? How can they be handled? Explain with an example.**
- 10. Explain the process of user authentication in Node.js with JWT.**



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Subject:- Full Stack Development

UNIT : 4

1. What is the benefit of using React in full stack development?
2. **Explain ReactDOM and its role in React applications.**
3. Define Components in React. Differentiate between functional and class-based components.
4. **Describe the React component lifecycle with a diagram.**
5. **Explain JS LocalStorage with an example of storing and retrieving data.**
6. Explain Composition and Inheritance in React with an example.
7. **Write a React component that fetches data using the Fetch API and displays it on the screen.**
8. What are React Events? Explain event handling in React with an example.



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Subject:- Full Stack Development

UNIT : 5

1. **Explain the different types of cloud service models (IaaS, PaaS, SaaS) with examples.**
2. **Compare major cloud providers like AWS, Azure, and Google Cloud.**
3. What is a Virtual Private Cloud? How is it different from a Public Cloud?
4. **Differentiate between horizontal and vertical scaling with examples.**
5. What is a Virtual Machine? Explain its types and use in cloud computing.
6. Explain the role of Ethernet in cloud networking.
7. What is a switch? How is it used in a virtualized network?
8. **What is Docker? Explain the architecture of Docker.**
9. **What is Kubernetes? Explain its architecture with a diagram.**